YOUR DIGITAL LAB

DECODE LABS REMOTE INTERNSHIP 2026

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- **INETRNSHIP DOMAIN:** DATA SCIENCE
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- **TASK #** 03(WEEK#03)

CUSTOMER SEGMENTATION USING PCA AND K-MEANS CLUSTERING

Abstract

Customer segmentation is an important business analytics technique used to identify groups of customers with similar behavior. This project applies Principal Component Analysis (PCA) and K-Means Clustering to segment mall customers into meaningful groups based on demographic and spending characteristics.

The objective was to discover hidden customer patterns without using labeled data. PCA was applied for dimensionality reduction, while K-Means clustering was used to identify customer groups. The Elbow Method and Silhouette Score were used to determine the optimal number of clusters. Finally, customer personas were created to support data-driven business decisions.

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1. Introduction

Businesses collect large amounts of customer data, but raw data alone does not provide actionable insights. Customer segmentation helps organizations understand different customer groups and design targeted marketing strategies.

In this project, an unsupervised machine learning approach was used to divide customers into distinct segments based on their purchasing behavior and demographic characteristics.

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2. Dataset Overview

The Mall Customers dataset was used for this analysis.

Dataset Information

Attribute	Value
Total Records	200
Features	5
Numerical Features	Age, Annual Income, Spending Score
Categorical Feature	Gender
Missing Values	0
Duplicate Records	0

Features Used

- Gender
- Age
- Annual Income
- Spending Score

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3. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the dataset using Pandas.
2. Checked dataset structure and data types.
3. Verified missing values.
4. Checked duplicate records.
5. Encoded Gender using Label Encoding.
6. Removed CustomerID column.
7. Applied StandardScaler for feature normalization.

Why Standardization?

Since Age, Income, and Spending Score are measured on different scales, StandardScaler was used to ensure that all features contribute equally during clustering.

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4. Dimensionality Reduction using PCA

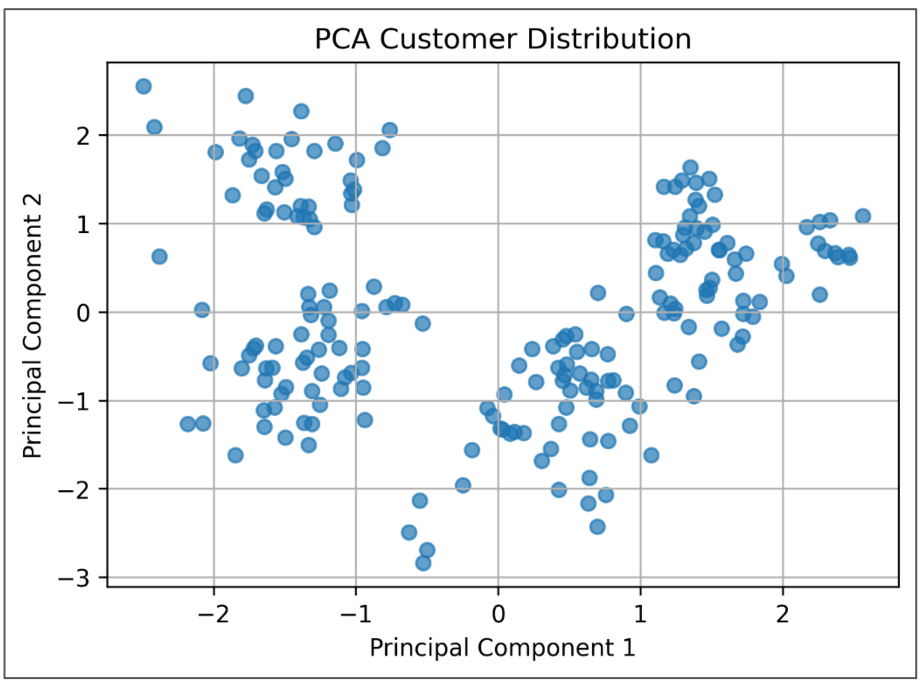
Principal Component Analysis (PCA) was applied to reduce the dataset from four dimensions to two dimensions.

Results

Metric	**Value**
Original Features	4
Principal Components	2
Variance Retained	~60%

PCA simplified the dataset while preserving most of the important information.

Figure 1: PCA Customer Distribution



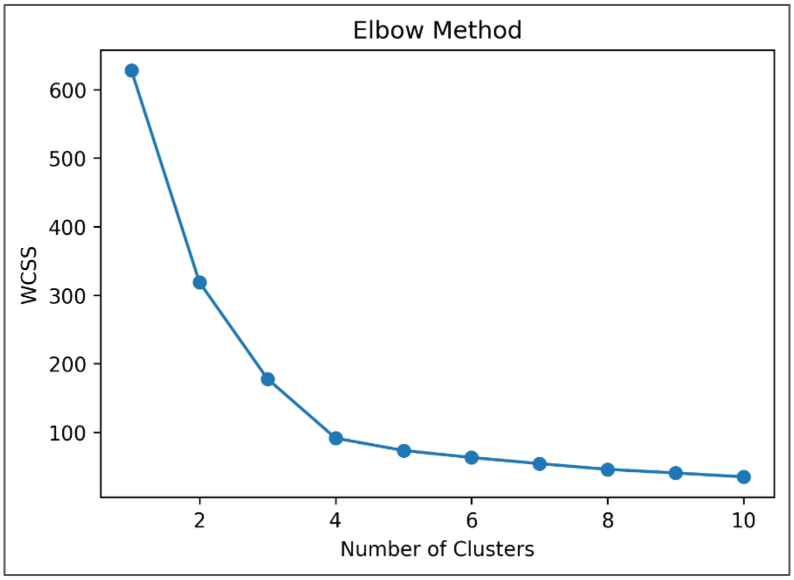
5. Optimal Cluster Selection

To determine the best number of clusters, two evaluation techniques were used.

5.1 Elbow Method

The Elbow Method measures Within Cluster Sum of Squares (WCSS). The optimal cluster count is identified at the point where additional clusters provide diminishing improvements.

Figure 2: Elbow Method



Observation

The elbow point appeared at K = 4.

5.2 Silhouette Score

The Silhouette Score evaluates how well each data point fits within its assigned cluster compared to other clusters.

Results

Highest Silhouette Score:

K = 4 → 0.4156

Observation

The Silhouette Score confirmed that four clusters provide the best balance between cohesion and separation.

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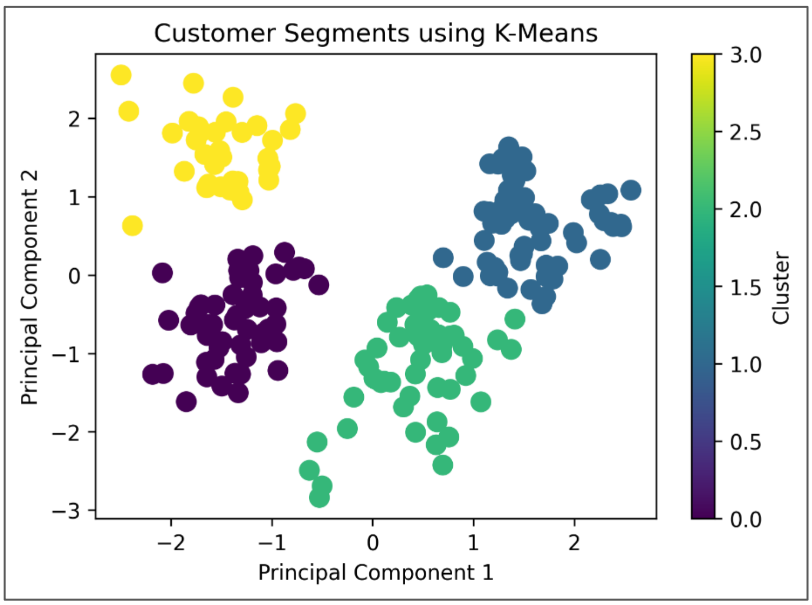
6. Customer Segmentation using K-Means

The K-Means algorithm was applied using K = 4.

Cluster Distribution

Cluster	Customers
0	34
1	51
2	54
3	61

Figure 3: Customer Segments using K-Means



Observation

The algorithm successfully separated customers into four visually distinct groups.

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7. Customer Personas

Cluster centroids were analyzed to create meaningful customer personas.

Cluster	Persona Name	Average Age	Average Income	Average Spending
0	High-Value Customers	29.91	81.50	73.88
1	Conservative Customers	47.65	74.63	30.86
2	Budget-Conscious Customers	49.22	46.22	34.00
3	Young Spenders	27.30	49.82	67.51

Business Recommendations

High-Value Customers

- Premium products
- Loyalty rewards
- Exclusive offers

Conservative Customers

- Retention campaigns
- Long-term membership programs

Budget-Conscious Customers

- Discounts and promotional offers
- Affordable product bundles

Young Spenders

- Social media marketing
- Trend-based promotions

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8. Interactive Dashboard

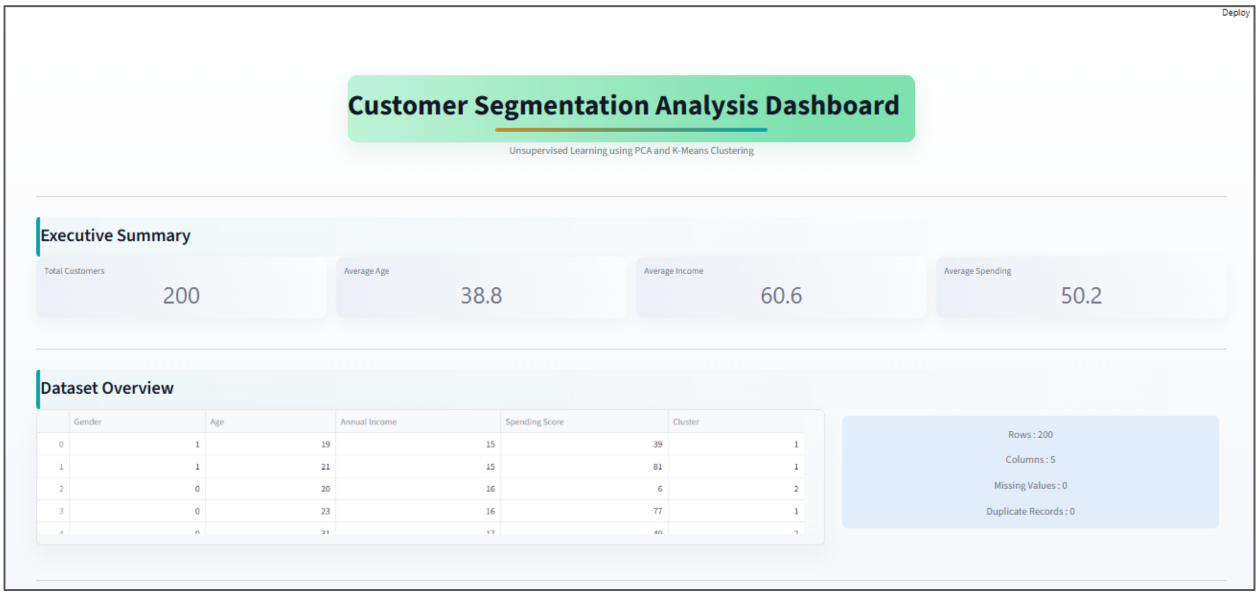
A professional Streamlit dashboard was developed to present project findings interactively.

Dashboard Features

- Executive Summary
- Dataset Overview

- Customer Demographics
- Correlation Analysis
- PCA Visualization
- Cluster Analysis
- Customer Personas
- Business Insights

Figure 4: Streamlit Dashboard



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9. Key Learnings

Through this project, the following concepts were learned and implemented:

- Unsupervised Learning
- Customer Segmentation
- Label Encoding
- StandardScaler
- Principal Component Analysis (PCA)
- K-Means Clustering
- Elbow Method
- Silhouette Score

- Cluster Interpretation
- Business Persona Creation
- Streamlit Dashboard Development

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10. Conclusion

This project successfully applied PCA and K-Means Clustering to identify meaningful customer segments from unlabeled retail data.

The Elbow Method and Silhouette Score both indicated that four clusters provide the most effective segmentation structure. These customer segments were translated into actionable business personas that can help organizations improve marketing strategies, customer engagement, and revenue generation.

The project demonstrates how unsupervised machine learning can transform raw customer data into valuable business intelligence.
